

1. Overview

The attendance system uses a school PC camera to recognize enrolled students and record attendance when a session is open. Two matchers are supported: LBPH (Local Binary Patterns Histogram) and ArcFace via OpenCV SFace with YuNet face detection. Both compare a live face to enrolled student photos, but they describe and compare faces differently.

2. Shared pipeline

Regardless of algorithm, the flow is:

- The camera captures a video frame.
- A face is detected in the frame.
- The face is compared to enrolled students.
- If the match is strong enough for several frames, and an attendance session is open, attendance is posted to the backend.

3. How LBPH works

LBPH describes a face using local texture patterns. The face is converted to grayscale. Each pixel is compared with nearby pixels to form a binary pattern, and those patterns become a histogram (a texture signature) for the student. A live face is matched by distance: lower distance means a better match.

In this project, LBPH accepts a match when the distance is at or below the configured threshold (about 70-75). The trained model is stored as `models/lbph.yml`.

Strengths:

- Simple and fast on a school PC.
- Small model and easy to run offline.
- Useful as a classic OpenCV baseline.

Weaknesses:

- Sensitive to lighting changes.
- Sensitive to pose, angle, and expression.
- Must be retrained when new students are added.
- More likely to confuse similar-looking students in uncontrolled conditions.

4. How ArcFace works (OpenCV SFace + YuNet)

ArcFace in this system means: detect the face, convert it into a numerical fingerprint (embedding), then compare that fingerprint to enrolled students. YuNet detects faces in the frame. SFace, trained with ArcFace-style learning, converts each face into an embedding vector. Similar faces produce similar vectors; different people produce different vectors.

During training (`train.py`), student photos become embeddings saved in `models/arcface_gallery.npz`. During live recognition, cosine similarity is used. Higher similarity means a stronger match. The system accepts matches around 0.36 and above (`ARCFACE_THRESHOLD`), then still requires consecutive frames, an open session, and a cooldown before recording attendance.

Strengths:

- More stable under lighting changes.
- Better with slight pose and angle variation.
- Better separation between different people.
- Closer to modern face recognition practice.

Weaknesses:

- Needs ONNX model files (YuNet and SFace).
- Slightly heavier than LBPH.
- Still needs clear enrollment photos and a gallery rebuild when adding students.

5. Comparison

Aspect	LBPH	ArcFace (SFace)
Method	Handcrafted texture patterns	Deep face embeddings
Comparison	Histogram distance	Cosine similarity
Lighting	Weak	Stronger
Pose / angle	Weak	Stronger
Accuracy	Good in controlled setup	Better overall
Speed	Very fast	Fast enough for school use
Model file	lbph.yml	arcface_gallery.npz
Best for	Simple lab baseline	Real hallway attendance

6. Why ArcFace is better for this project

A school hallway is not a controlled lab. Lighting changes, students move, and faces are not always frontal. ArcFace is the better production matcher because:

- It learns identity features instead of relying only on local pixel texture.
- It handles lighting and pose changes more reliably with a Tapo camera.
- It reduces false matches between different students.
- It reflects modern face recognition practice, which strengthens the capstone.
- For day-to-day attendance use, it is more reliable than LBPH.

LBPH remains useful as a baseline algorithm. ArcFace is the preferred default matcher for real school attendance recognition.

7. One-sentence summary

LBPH compares face textures; ArcFace compares learned face embeddings. ArcFace is better because those embeddings stay more stable when lighting, angle, and camera conditions change.